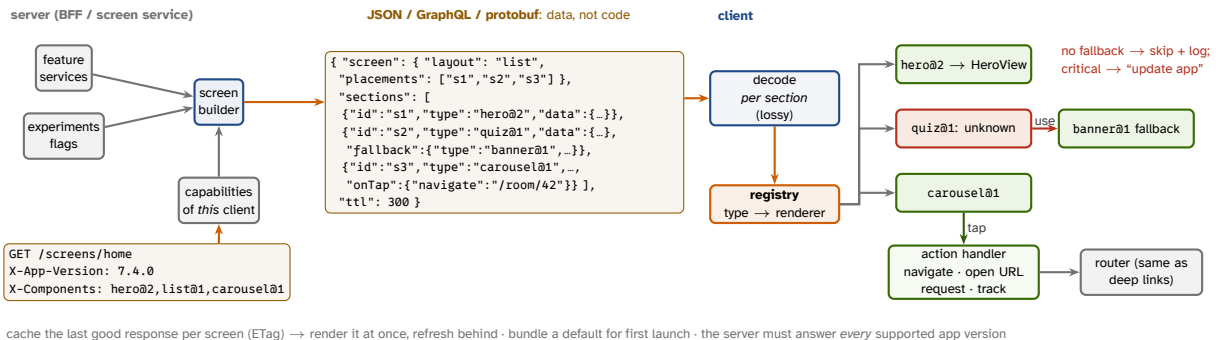


**In one line:** The server sends a typed **description** of a screen — *layout + components with their data + actions* — and the app renders it with **native components it already ships**. You gain release independence (change a screen without a store release, run experiments, one backend for iOS/Android/web); you pay with a schema to design, **every old app version** to keep working, and a weaker compile-time contract. It moves **data and composition**, never code.



## How it works

- How much is driven?** A spectrum: *flags/config* → *content* (copy, images) → **composition** (which known sections, in what order, with what data) → *primitives* (stack/text/image = a browser). Most production systems sit at composition of *product-level* sections; primitives rebuild HTML with worse tooling.
- Schema:** a typed contract with a **type discriminator** per component (GraphQL union/interface, protobuf oneof, JSON + JSON Schema); models **codegen'd** for each client. Data arrives *ready to show* — translated, localised, formatted — so clients stay dumb.
- Layout vs content:** the screen says where things go (placements, list/grid, modal vs push); a *section* carries only its data and knows nothing of the screen, so it can be reused on any screen.
- Registry:** type → renderer; each renderer decodes *its own* payload, so one malformed section cannot blank the screen.
- Versioning:** a type's shape is frozen once shipped; a breaking change is a **new type** (hero@2). Only add optional fields. Old app versions live for years — the server must serve all of them.
- Capability negotiation:** the client sends app version + the component types/versions it renders; the server emits only those, or attaches a fallback. The client still defends itself: unknown type → fallback → skip + log; unknown action / enum case → no-op default.
- Actions** are *declarative intents*, not code: `navigate(route)`, `openURL`, `request(endpoint, then:)`, `track(event)`, `dismiss`, `sequence`. The client owns execution; routes go through the **same router as universal links**. Local UI state (toggles, text fields) stays on the client; the server re-validates.
- Cache + offline:** store the last good response per screen, render it immediately, refresh behind (stale-while-revalidate); ship a bundled default.
- Performance:** a round trip before first paint → prefetch, cache, skeletons, paginate sections, stable ids for diffing, decode off main. Big payloads were a named pain point at Airbnb.
- Testing:** snapshot every component × fixtures (incl. long text, RTL, Dynamic Type); contract tests on the schema; a **compat matrix** (oldest supported app × new server response); a preview tool for authors.
- Accessibility:** native components keep VoiceOver, Dynamic Type, RTL — a win over web views; the server must still send labels and semantics (heading, button).

## When NOT to

Bespoke, animated, gesture-heavy UI (editors, camera, games — coordinated animations were a named limit at Lyft); offline-first core flows; small teams (high upfront cost); a latency-critical first screen with no cache; when a web view would do.

## Example — a defensive registry

```
struct Section: Decodable { let id: String; let type: String
  let data: JSONValue; let fallback: [Section]? }
protocol Renderer { static var type: String { get } // "hero@2"
  func view(_ data: JSONValue, _ on: ActionHandler) throws -> AnyView }
@MainActor final class Registry {
  var renderers: [String: any Renderer] = [:]
  var supported: String { renderers.keys.sorted().joined(separator: ",") }
  func render(_ s: Section, _ on: ActionHandler) -> AnyView? {
    if let r = renderers[s.type], let v = try? r.view(s.data, on) { return v }
    if let f = s.fallback?.first { return render(f, on) } // server's plan B
    on.log(.unrendered(s.type, s.id)); return nil // skip, never crash
  }
}
enum Action: Decodable { case navigate(String), openURL(URL),
  track(String), unknown } // custom init: unknown keys -> .unknown
```

supported goes into the capabilities header; the list of sections is decoded as [Section] with data kept raw, so a bad section fails alone.

## App Review — what Apple actually says

- 2.5.2:** apps “may not ... download, install, or execute code which introduces or changes features or functionality of the app”. SDUI ships *data* rendered by reviewed native code — that is the defensible line.
- 2.3.1(a):** no “hidden, dormant, or undocumented features”; new features described in Review Notes — so a server flag must not switch on something App Review never saw.
- 2.1(a):** backend on, demo account ready — review sees the server's output.
- The license agreement's executable-code clause allows *interpreted* code only if it does not change the app's primary purpose (the basis for JS OTA updates, not SDUI).

## Traps

- ! “SDUI skips App Review” — no: it recombines *reviewed* features.
- ! Changing a shipped type's payload in place → old clients break.
- ! Strict decode of the whole screen: one unknown case = empty screen.
- ! Going down to primitives: a browser without the browser's tooling.
- ! No cache → spinner on every launch; testing only the newest client.

## Remember

**Server says *what*, client owns *how*.** Freeze types, add versions, negotiate capabilities, fall back, never crash.